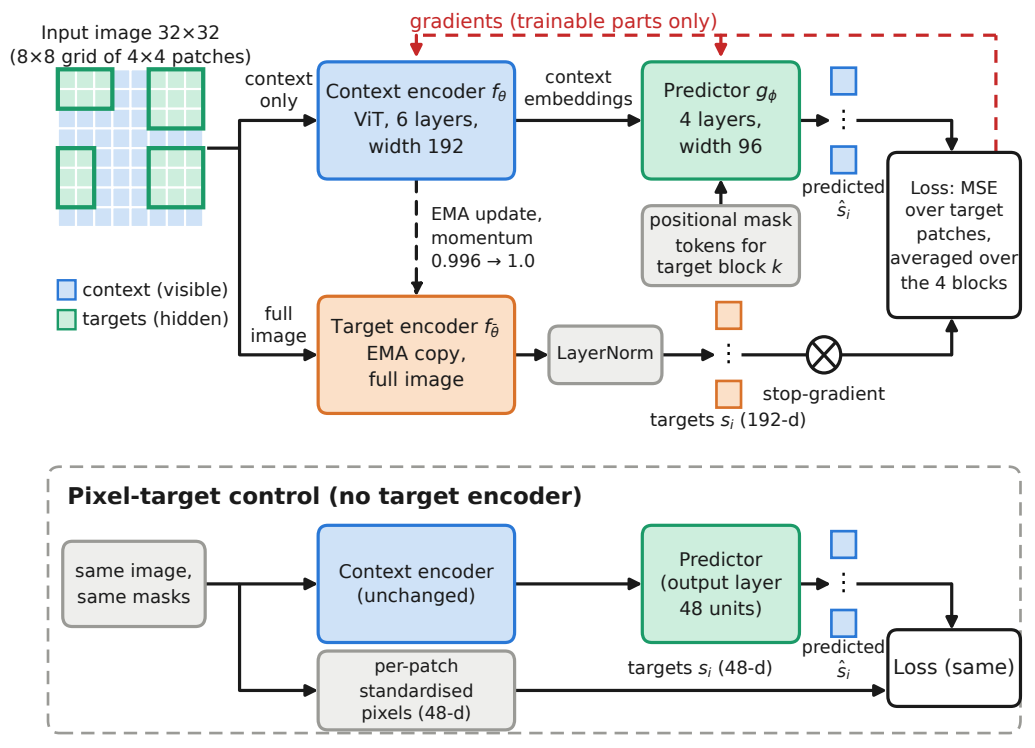
gradients (trainable parts only)
Input image 32×32
(8×8 grid of 4×4 patches)
context only
Context encoder f_θ
ViT, 6 layers,
width 192
context embeddings
Predictor g_ϕ
4 layers,
width 96
predicted $\hat{s}_i$
Loss: MSE over target patches, averaged over the 4 blocks
EMA update,
momentum
0.996 → 1.0
positional mask tokens for target block k
context (visible)
targets (hidden)
full image
Target encoder $f_{\bar{\theta}}$
EMA copy,
full image
LayerNorm
stop-gradient
targets s_i (192-d)
Pixel-target control (no target encoder)
same image,
same masks
Context encoder
(unchanged)
Predictor
(output layer
48 units)
predicted $\hat{s}_i$
Loss (same)
per-patch
standardised
pixels (48-d)
targets s_i (48-d)